The Rising Sun: Shopping List

For the first prototype, the light only. Written 2026-10-18 for Dr. O.

No prices are listed because I cannot check them from here. Nothing below has been tried on real parts. If a listing does not match what is written, do not buy it, and ask first.

Buy for the first prototype

- Controller board: you already have the ESP32-S3 N16R8 from the Holopod build. Use one of those. Nothing to buy.
- LED strip: search "WS2812B LED strip 5V". It must say 5V, WS2812B (or WS2812), and "individually addressable". Buy 1 meter with 60 LEDs per meter. Black or white backing both work. Do NOT buy 12V strips, or strips that say "RGB" with only 4 wires and no data line.
- Power supply: a 5V supply rated 4 amps or more, with a screw terminal or a barrel plug and a matching adapter. 60 LEDs can draw up to 3.6 amps at full white. Do NOT use a phone charger.
- Resistor: 330 ohm, any small assortment kit has it.
- Capacitor: 1000 microfarad, 6.3V or higher. Also in most kits.
- Jumper wires: male to female and male to male, a small assortment.
- Breadboard or a few screw terminal blocks, so nothing has to be soldered for the first try.
- USB cable: it must carry data, not just power. If the board does not show up on the computer, a charge-only cable is the usual reason.

Buy only if the first light flickers

- A level shifter, 74AHCT125 (a small board, sold as "logic level shifter"). The board sends a 3.3V signal and some strips want 5V.

Do not buy yet

- The full length of strip for the acrylic. How many LEDs it needs depends on the acrylic shape, which is not decided. Buy it after David has seen the prototype and you have picked the layout.
- The bigger power supply. It is sized from the final LED count: about 60 milliamps per LED at full white, and a safe supply has some room above that.
- Her voice parts (speaker, amplifier, microphone). Not built yet. When they are wanted, the likely parts for this board are a small I2S amplifier (MAX98357A) with a speaker and an I2S microphone (INMP441). That is from memory, not checked against a build.

Before David starts

- Lay the parts out and tick them off against this list.
- Give him the First Prototype sheet. It lists every step.
- He does not need any account, login or key for the first prototype.
- Ask him to send you a photo of the strip lit. That is the only report you need today.